The main purpose of this document is to offer some additional ideas, based on the framework of [2]. These ideas are along the line of imposing additional assumptions so that we can have honest and adaptive pointwise confidence intervals of KRR.

Using notations in [2], let $\widetilde{f}_i = L_{K^{1/2}} f_i$ be the normalized input function, $\widetilde{\beta}_0 = L_{K^{-1/2}}\beta_0$ be the normalized regression functional. Let $T = \mathbb{E}[\widetilde{f}_i \otimes \widetilde{f}_i]$ admit the spectrum decomposition $T = \sum_{j=1}^{\infty} s_j \psi_j \otimes \psi_j$ with $s_j \asymp j^{-2r}$ and $\{\psi_j\}$ being an orthonormal basis of $L^2(\mathcal{X})$.

Suppose $\widetilde{\beta}_0 = \sum_j \vartheta_j \psi_j$. We only know $\sum_j \vartheta_j^2 \le 1$ and $\mathbb{E}[\alpha_j^2] = s_j$, but nothing else. Given a test function v with $\widetilde{v} = L_{K^{1/2}} v = \sum_j \alpha_j \psi_j$, such that $\sum_j \alpha_j^2 \le 1$. If we know T (and therefore $\{\psi_j\}$), we would know $\{\alpha_j\}$. Roughly speaking, omitting supposedly higher-order terms, the estimation error can be decomposed into

$$\left|\langle \widetilde{v}, \lambda(T+\lambda I)^{-1}\widetilde{\beta}_0\rangle\right| + \frac{1}{n}\left|\sum_{i=1}^{n} \varepsilon_i \langle \widetilde{v}, (T_n+\lambda I)^{-1}\widetilde{f}_i\rangle\right|.$$

Consequently, the *squared* estimation error on $\widetilde{v}$ is upper bounded by

$$\underbrace{\left|\sum_{j=1}^{\infty} \frac{\lambda \vartheta_j \alpha_j}{s_j+\lambda}\right|^2}_{=:\text{bias}^2} + \underbrace{\frac{1}{n}\sum_{j=1}^{\infty} \frac{\alpha_j^2}{s_j+\lambda}}_{=:\text{variance}}.$$

Here in the variance term we transit from T_n to T to avoid discussion of empirical covariance eigenvalues/eigenfunctions.

We seek a CI function $\Delta(\cdot)$ that satisfies the following properties:

1. $\Delta(\cdot)$ does not depend on $\{\vartheta_j\}$;
2. With high probability, for any $\widetilde{v}$, $\Delta(\widetilde{v})^2$ upper bounds the squared error at $\widetilde{v}$;
3. $\mathbb{E}[\Delta(\cdot)^2]$ is roughly on the right order.

0.1 What is the right order?

On the bias side, we have (note this does *not* require any assumptions on $\{s_j\}$ at all.)

$$\mathbb{E}[\text{bias}^2] = \sum_{j=1}^{\infty} \frac{\lambda^2 \vartheta_j^2 s_j}{(s_j+\lambda)^2} \le \left(\sum_{j=1}^{\infty} \vartheta_j^2\right) \sup_{s>0} \frac{\lambda^2 s}{(s+\lambda)^2} \lesssim \lambda.$$

Note that the first equality holds because $\mathbb{E}[\alpha_j \alpha_k] = \delta_{jk} s_j$ because $\{\psi_j\}_{j=1}^{\infty}$ is an ONB for T with eigenvalues $\{s_j\}_{j=1}^{\infty}$.

On the variance side, with $s_j \lesssim j^{-2r}$, we have

$$\mathbb{E}[\text{variance}] = \frac{1}{n}\sum_{j=1}^{\infty} \frac{s_j}{s_j+\lambda} \lesssim \frac{1}{\lambda^{1/2r} n}.$$

So we should aim for

$$\mathbb{E}[\Delta(\widetilde{v})^2] \lesssim \lambda + \frac{1}{\lambda^{1/2r} n} \lesssim n^{-2r/(2r+1)}$$

for $\lambda \asymp n^{-2r/(2r+1)}$.

0.2 A simple idea

By Cauchy-Schwarz inequality, it holds that

$$\text{bias}(\widetilde{v})^2 \leq \left(\sum_{j=1}^{\infty} \vartheta_j^2\right) \cdot \lambda^2 \left(\sum_{j=1}^{\infty} \frac{\alpha_j^2}{(s_j+\lambda)^2}\right) \lesssim \lambda^2 \left(\sum_{j=1}^{\infty} \frac{\alpha_j^2}{(s_j+\lambda)^2}\right),$$

where $\alpha_j = \langle \widetilde{v}, \psi_j \rangle$ is known. This leads to

$$\mathbb{E}[\Delta(\widetilde{v})^2] \lesssim \lambda^2 \left(\sum_{j=1}^{\infty} \frac{s_j}{(s_j+\lambda)^2}\right) \lesssim \lambda^{1-1/2r},$$

which is losing a factor of $\lambda^{1/2r}$.

1 Constraints on regression functional: anisotropic krr

We consider using *changing* regularization parameters so that $T + \lambda I$ is replaced with $T + \Lambda$ where $\Lambda = \sum_{j=1}^{\infty} \lambda_j \psi_j \otimes \psi_j$. We also denote $\lambda_0 \asymp n^{-2r/(2r+1)}$ for reference purposes. Then at any $\widetilde{v} = \sum_j \alpha_j \psi_j$,

$$\text{bias}(\widetilde{v})^2 + \text{variance}(\widetilde{v}) \lesssim \left|\sum_{j=1}^{\infty} \frac{\lambda_j \vartheta_j \alpha_j}{s_j + \lambda_j}\right|^2 + \frac{1}{n} \sum_{j=1}^{\infty} \frac{\alpha_j^2}{s_j + \lambda_j},$$

because the bias term is now $(T+\Lambda)^{-1}\Lambda\widetilde{\beta}_0$.

We are going to use

$$\lambda_j := s_j \sqrt{\lambda_0}/|\alpha_j|.$$

The intuition is to upper bound the bias term in a uniform manner:

$$\left|\frac{\lambda_j \alpha_j}{s_j + \lambda_j}\right| \leq \frac{\sqrt{\lambda_0}}{1 + \sqrt{\lambda_0/|\alpha_j|}} \leq \sqrt{\lambda_0}, \qquad \forall j \geq 1.$$

1.1 Assumptions needed

Due to existence of uncorrelated but non-zero cross-terms in the bias-square expression, some restrictions on $\widetilde{\beta}_0 = \sum_{j=1}^{\infty} \vartheta_j \psi_j$ must be imposed, in addition to $\sum_j \vartheta_j^2 \leq 1$ implied by square integrability of $\widetilde{\beta}_0$. Both assumptions below must be imposed with respect to $\{\psi_j\}_{j=1}^{\infty}$; that is, for any policy (or near-optimal policy) the bandit algorithm constructs.

Option 1. We assume $\sum_{j=1}^{\infty} |\vartheta_j| \leq M < +\infty$. This implies

$$\left|\sum_{j=1}^{\infty} \frac{\lambda_j \vartheta_j \alpha_j}{s_j + \lambda_j}\right| \leq M \sup_{j \geq 1} \frac{\lambda_j |\alpha_j|}{s_j + \lambda_j} \leq M\sqrt{\lambda_0}.$$

Remark 1.1. This assumption is implied by the decay condition $\vartheta_j \lesssim j^{-\beta}$ for any $\beta > 1$, which is imposed and necessary for fPCA analysis (see e.g. [1]). Note that fPCA analysis also requires this specific coefficient decay on the basis $\{\psi_j\}$, which is not basis independent.

Remark 1.2. Under the context of fPCA analysis in [1], $|\alpha_j| \lesssim j^{-\gamma}$ and $|\beta_j| \lesssim j^{-\beta}$. Assuming we use an RKHS with spectral decay $j^{-(\beta-1-\epsilon)}$ and T and C commute, we have $r = 2\gamma+\beta-1-\epsilon \approx 2\gamma+\beta-1$ and the MSE rate is $n^{-2r/(2r+1)} = n^{-(4\gamma+2\beta-2)/(4\gamma+2\beta-1)} = n^{-1+\frac{1}{4\gamma+2\beta-1}}$. On the other hand, with the understanding of $\alpha = 2\gamma$, the fPCA MSE rate is $n^{-2(\beta+\gamma-1)/(\alpha+2\beta-1)} = n^{-(2\beta+2\gamma-2)/(2\beta+2\gamma-1)} = n^{-1+\frac{1}{2\beta+2\gamma-1}}$. fPCA remains sub-optimal.

Remark 1.3. If $\{\psi_j\}$ is the Fourier basis, $\sum_{j=1}^{\infty} |\vartheta_j| < +\infty$ implies $\widetilde{\beta}_0$ is absolutely continuous. (On the other hand, if $\widetilde{\theta}_0$ is Hölder continuous in any $\alpha > 0$, then $\sum_{j=1}^{\infty} |\vartheta_j| < +\infty$ on the Fourier basis.) If $\{\psi_j\}$ is the wavelet basis, it implies $\widetilde{\beta}_0 \in B^{1/2}_{1,1}$. These are *some* regularity conditions, but not super strong. The stronger assumption of $|\vartheta_j| \lesssim j^{-\beta}$ for $\beta > 1$, in either the Fourier or the wavelet basis, implies that $\widetilde{\beta}_0$ is in the fractional sobolev space H^s with $s = 1/2$. This is a tiny amount of smoothness that we'll not be able to capture, potentially leading to some sub-optimality.

Option 2. We assume $\{\vartheta_j\}_{j=1}^{\infty}$ is random. Furthermore, $\vartheta = (\vartheta_j)_{j=1}^{\infty}$ is centered, sub-Gaussian, with variance proxy Σ that satisfies the following properties:

1. $\mathrm{tr}(\Sigma) = \sum_{j=1}^{\infty} \Sigma_{jj} \le M < +\infty$, mimicking square integrability of $\widetilde{\beta}_0$;
2. Σ is *"weakly" diagonally dominant*, meaning that there exists $C < +\infty$ such that for all $j \ge 1$, $\sum_{k \ne j} |\Sigma_{jk}| \le C \cdot \Sigma_{jj}$.

In this case, using the celebrated condition that $|ab| \le \frac{1}{2}(a^2+b^2)$ for any $a, b \in \mathbb{R}$, we obtain

$$\mathbb{E}_{\vartheta}\left[\left(\sum_{j=1}^{\infty} \frac{\lambda_j \vartheta_j \alpha_j}{s_j+\lambda_j}\right)^2\right] = \sum_{j,k=1}^{\infty} \frac{\Sigma_{jk}\lambda_j\lambda_k\alpha_j\alpha_k}{(s_j+\lambda_j)(s_k+\lambda_k)} \overset{(a)}{\le} (C+1)\cdot \sum_{j=1}^{\infty} \frac{\Sigma_{jj}\lambda_j^2\alpha_j^2}{(s_j+\lambda_j)^2}$$
$$\le (C+1)M \cdot \sup_{j>0} \frac{\lambda_j^2\alpha_j^2}{(s_j+\lambda_j)^2} \le (C+1)M \cdot \lambda_0.$$

Note that all expectations here are taken with respect to ϑ, fixing $\widetilde{v}$ and $\{\alpha_j\}$. In Eq. (a) we used weakly diagonal dominance of M.

With sub-gaussianity, with high probability (over the randomness in $\widetilde{\beta}_0$) the (squared) bias is upper bounded by $\lambda_0 \cdot O(\log n)$.

Remark 1.4. If $\widetilde{\beta}_0$ is samples from a Gaussian Process (GP) on $[0,1]$, then by the Kosambi-Karhunen-Loeve theorem, there exists an orthonormal basis $\{\phi_j\}_{j=1}^{\infty}$ of $L^2([0,1])$ such that $\widetilde{\beta}_0 = \sum_{j=1}^{\infty} Z_j\phi_j$, where $\{Z_j\}$ are centered, independent Normal random variables with $\mathbb{E}[Z_j^2] = \mu_j$. If T and K commute (i.e. $\psi_j \equiv \phi_j$), then the assumption is satisfied. If exact commutability is not available, then $Z \sim \mathcal{N}(0, WDW^*)$ where $W_{ij} = \langle \phi_i, \psi_j\rangle$ and $D = \mathrm{diag}(\mu_1, \mu_2, \cdots)$, and the assumption is implied by

$$C \times \left(\sum_{\ell=1}^{\infty} \mu_\ell \langle \phi_j, \psi_\ell\rangle^2\right) \;\ge\; \sum_{k \ne j} \left|\sum_{\ell=1}^{\infty} \mu_\ell \langle \phi_j, \psi_\ell\rangle\langle \phi_k, \psi_\ell\rangle\right|, \qquad \forall j = 1, 2, \cdots$$

1.2 The variance term

For the variance term, we have

$$\frac{1}{n}\sum_{j=1}^{\infty} \mathbb{E}\left[\frac{\alpha_j^2}{s_j+\lambda_j}\right] \le \frac{1}{n}\sum_{j=1}^{\infty} \mathbb{E}\left[\frac{\alpha_j^2}{s_j(1+\sqrt{\lambda_0}/|\alpha_j|)}\right].$$

We have

$$\mathbb{E}\left[\frac{\alpha_j^3}{|\alpha_j|+\sqrt{\lambda_0}}\right] = \mathbb{E}\left[\alpha_j^2 \cdot \frac{|\alpha_j|}{|\alpha_j|+\sqrt{\lambda_0}}\right] \overset{(a)}{\leq} \sqrt{\mathbb{E}[\alpha_j^4]}\sqrt{\mathbb{E}\left[\frac{\alpha_j^2}{(|\alpha_j+\sqrt{\lambda_0})^2}\right]}$$
$$\overset{(b)}{\leq} \sqrt{\mathbb{E}[\alpha_j^4]}\sqrt{\mathbb{E}\left[\frac{\alpha_j^2}{\alpha_j^2+\lambda_0}\right]} \overset{(c)}{\leq} \sqrt{\mathbb{E}[\alpha_j^4]}\sqrt{\frac{\mathbb{E}[\alpha_j^2]}{\mathbb{E}[\alpha_j^2]+\lambda_0}} \leq \sqrt{\mathbb{E}[\alpha_j^4]}\sqrt{\frac{s_j}{s_j+\lambda_0}},$$

where Eq. (a) holds by Cauchy-Schwarz inequality, Eq. (b) holds because $(a+b)^2 \geq |a|^2+|b|^2$, and Eq. (c) holds by Jensen's inequality and the fact that $f(x) = x/(a+x)$ is concave on $a > 0$ and $x > 0$. Under the condition that $\mathbb{E}[\alpha_j^4] \lesssim (\mathbb{E}[\alpha_j^2])^2 \lesssim s_j^2$, we have

$$\mathbb{E}\left[\frac{\alpha_j^3}{s_j(|\alpha_j|+\sqrt{\lambda_0})}\right] \lesssim \frac{\sqrt{s_j}}{\sqrt{s_j+\lambda_0}}.$$

1.3 Without assuming known T

Suppose $T = \sum_{j=1}^{\infty} s_j \psi_j \otimes \psi_j$ and $T_n = \sum_{j=1}^{n} \widehat{s}_j \widehat{\psi}_j \otimes \widehat{\psi}_j$. Because there is no eigengap on small directions, it is essentially not possible to upper bound the difference between ψ_j and $\widehat{\psi}_j$ for large j. Because also ℓ_1-summability is not basis independent, we have no choice but to assume the following:

$$\sum_{j=1}^{n} |\widehat{\vartheta}_j| \leq O(1), \qquad \text{where } \widehat{\vartheta}_j = \langle \widetilde{\beta}_0, \widehat{\psi}_j \rangle.$$

The anisotropic KRR solution is on a given function f is then

$$\langle (T_n + \Lambda_n)^{-1} \frac{1}{n} \sum_{i=1}^{n} y_i L_{K^{1/2}} f_i, L_{K^{1/2}} f \rangle,$$

where

$$\Lambda_n = \sum_{j=1}^{n} \widehat{\lambda}_j \widehat{\psi}_j \otimes \widehat{\psi}_j; \qquad \widehat{\lambda}_j = \frac{\widehat{s}_j \sqrt{\lambda_0}}{|\widehat{\alpha}_j|} = \frac{\widehat{s}_j \sqrt{\lambda_0}}{|\langle L_{K^{1/2}} f, \widehat{\psi}_j \rangle|}.$$

The bias term is unchanged, because on the particular input function, the bias is

$$\left| \sum_{j=1}^{\infty} \frac{\widehat{\lambda}_j \widehat{\vartheta}_j \widehat{\alpha}_j}{\widehat{s}_j + \widehat{\lambda}_j} \right|$$

and with everything replaced by the empirical version, the Hölder's inequality and sup-argument remains the same.

What changes is the out-of-sample variance. The variance is upper bounded by

$$\frac{1}{n} \sum_{j=1}^{n} \mathbb{E}\left[\frac{\widehat{\alpha}_j^2}{\widehat{s}_j(1+\sqrt{\lambda_0}/|\widehat{\alpha}_j|)}\right]$$

where expectation is taken over the population distribution. Using the same analysis, it is upper bounded by

$$\sqrt{\mathbb{E}[\widehat{\alpha}_j^4]}\sqrt{\frac{\mathbb{E}[\widehat{\alpha}_j^2]}{\mathbb{E}[\widehat{\alpha}_j^2]+\lambda_0}}$$

We need to show, roughly speaking, the following (with $v = L_{K^{1/2}} f$ and $v_i = L_{K^{1/2}} f_i$)

$$\mathbb{E}[\widehat{\alpha}_j^2] = \mathbb{E}[\langle v, \widehat{\psi}_j \rangle^2] \overset{??}{\leq} O(1) \cdot \left[\frac{1}{n} \sum_{i=1}^{n} \langle v_i, \widehat{\psi}_j \rangle^2 \right] = \widehat{s}_j.$$

For fixed $\widehat{\psi}_j$ this is concentration inequality but we need to be uniform over all $\widehat{\psi}_j$ - not sure if it can be dimension independent. We need similar things for the 4th moment too.

2 Constraints on input function: uniform CI

We now consider a set of alternative assumptions placed on $\widetilde{v} = L_{K^{1/2}} f$. When appropriate, these conditions would lead to *uniform* error bounds for almost every $\widetilde{v}$.

2.1 Assumption needed

We impose the following assumptions on the distribution of $\widetilde{v}$:

1. There exists a psd operator V such that $\widetilde{v}$ is centered and sub-Gaussian with variance proxy V; that is, for any $\phi \in L^2([0,1])$, $\mathbb{E}[e^{\langle \phi, \widetilde{v} \rangle}] \leq \exp\{\langle \phi, V\phi \rangle / 2\}$.
2. For any action $a \in \mathcal{A}$ and regularization parameter $\lambda \asymp n^{-2r/(2r+1)}$, let $T_\lambda^a := \mathbb{E}[\widetilde{f} \otimes \widetilde{f} | a^*(\widetilde{f}) = a] + \lambda I$ be the smoothed population covariance operator on the region where a is optimal. Then $\|(T_\lambda^a)^{-1/2} V (T_\lambda^a)\|_{\text{op}} \leq O(1)$.

This is a "spectral alignment" condition that very similarly resemble $\lambda_{\min}$ conditions in margin-based contextual bandit; e.g. [3].

2.2 Analysis on bias

On the bias side, assuming the optimal action is never eliminated, for action a it holds that

$$\mathbb{E}[\langle \widetilde{v}, \widetilde{\beta}_\lambda^a - \widetilde{\beta}_0^a \rangle^2] \leq \lambda^2 K \mathbb{E}[\langle \widetilde{v}, (T_\lambda^a)^{-1} \widetilde{\beta}_0 \rangle^2] \leq \lambda K \|\widetilde{\beta}_0\|_2^2 \cdot \|(T_\lambda^a)^{-1/2} V (T_\lambda^a)\|_{\text{op}} \lesssim O(\lambda).$$

Because $\widetilde{v}$ is sub-Gaussian with a matrix variance proxy, this ensures that the bias for $\widetilde{v}$ is upper bounded above with high probability. This means, with high probability, $\widetilde{v}$ has the same '

2.3 Analysis on variance

We can always use pointwise CI because we know variance. If we seek a uniform bound for almost all $\widetilde{v}$ on this term too, we need the following stronger condition:

$$\|(T^a)^{-1/2} V (T^a)^{-1/2}\|_{\text{op}} \leq O(1).$$

This implies $\langle \phi, T_a \phi \rangle \leq O(1) \cdot \langle \phi, V\phi \rangle$ for any $\phi \in L^2([0,1]$. We then have

$$\begin{aligned} \mathbb{E}[\langle \widetilde{v}, (T + \lambda I)^{-1} v \rangle] &\lesssim K \cdot \mathbb{E}[\langle \widetilde{v}, (T_\lambda^a)^{-1} \widetilde{v} \rangle] = K \cdot \text{tr}((T_\lambda^a)^{-1/2} V (T_\lambda^a)) \\ &\leq K \text{tr}((T_\lambda^a)^{-1/2} T^a (T_\lambda^a)^{-1/2}) = K \cdot \sum_{j=1}^{\infty} \frac{s_j}{s_j + \lambda}. \end{aligned}$$

2.4 Without assuming known T

By using the Berstein that considers the effective dimension like [6], we can derive

$$\left\| T_\lambda^{-1/2} (T - \widehat{T}) T_\lambda^{-1/2} \right\|_{\text{op}} \leq \frac{1}{2}$$

with high probabilty, which further imples

$$\frac{1}{2}T_\lambda \preceq \widehat{T}_\lambda \preceq \frac{3}{2}T_\lambda$$

w.h.p.. Then, for the variance of the bias term

$$\langle L_{\boldsymbol{K}^{1/2}}x, \lambda\widehat{\boldsymbol{T}}_\lambda^{-1}f_0\rangle,$$

we have

$$\begin{aligned}\mathbb{E}\left[\langle L_{\boldsymbol{K}^{1/2}}x, \lambda\widehat{\boldsymbol{T}}_\lambda^{-1}f_0\rangle^2\right] = \lambda^2\langle f_0, \widehat{T}_\lambda^{-1}V\widehat{T}_\lambda^{-1}f_0\rangle &\leq \lambda^2\left\|f_0\right\|^2\left\|\widehat{T}_\lambda^{-1}V\widehat{T}_\lambda^{-1}\right\|_{\mathrm{op}}\\ &\lesssim \lambda^2\left\|f_0\right\|^2\left\|\widehat{T}_\lambda^{-1}T\widehat{T}_\lambda^{-1}\right\|_{\mathrm{op}}\\ &\lesssim \lambda^2\left(\left\|\widehat{T}_\lambda^{-1}\widehat{T}\widehat{T}_\lambda^{-1}\right\|_{\mathrm{op}} + \left\|\widehat{T}_\lambda^{-1}(T-\widehat{T})\widehat{T}_\lambda^{-1}\right\|_{\mathrm{op}}\right)\\ &\lesssim \lambda^2(\lambda^{-1}+\frac{1}{2}\lambda^{-1}) \sim \lambda,\end{aligned}$$

which is the right order. For the variance term itself we have

$$\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}g_\epsilon\rangle = \frac{1}{n}\sum_{i=1}^n \varepsilon_i\langle L_{\boldsymbol{K}^{1/2}}x, \widehat{\boldsymbol{T}}_\lambda^{-1}L_{\boldsymbol{K}^{1/2}}x_i\rangle,$$

whose variance is

$$\frac{\sigma^2}{n}\sum_{i=1}^n \mathrm{Tr}\left(\widehat{T}_\lambda^{-1}\Sigma\widehat{T}_\lambda^{-1}\widehat{T}\right) \lesssim \frac{1}{n}\mathrm{Tr}\left(\widehat{T}_\lambda^{-1}\Sigma\right) \lesssim \frac{1}{n}\mathrm{Tr}\left(\widehat{T}_\lambda^{-1}T\right) \lesssim \frac{1}{n}\mathrm{Tr}\left(T_\lambda^{-1}T\right) \sim \frac{\lambda^{-1/(2r)}}{n},$$

which is also of the right order.

3 Optimality of CI

Let $\Delta(\cdot)$ be a CI function associated with an estimator $\widehat{\theta}$. We say that $\Delta(\cdot)$ is an *honest and adaptive* confidence interval if the following holds:

1. For some type-I error bound $\epsilon \in (0,1)$, it holds for (almost) all f that $\Pr[|\langle f, \widehat{\beta}-\beta_0\rangle| \leq \Delta(f)] \geq 1-\epsilon$, where randomness is over $\{x_i, \varepsilon_i\}$.
2. $\mathbb{E}[\Delta(f)^2]$ is on the same order of $\mathbb{E}[\langle f, \widehat{\beta}-\beta_0\rangle^2]$, which should match the minimax rate (e.g. $n^{-2r/(2r+1)}$).

Note that the first requirement rules out high-probability CIs that are sufficient for bandit purposes, such as methods/analysis in Sec. 2. Nevertheless, for distributions of f that are not sub-Gaussian, it might be possible to relax the first requirement to hold for a constant fraction of f while still getting the required lower bound.

3.1 Summary of Prediction lower bound

This is a high-level summary of minimax prediction lower bound established in [2]. For simplicity, we use the identity kernel, while general kernels can be handled by simply applying the $L_{K^{1/2}}$ operator. For notational simplicity, we will use $\beta_0 = \sum_{j=1}^\infty \theta_j\psi_j$ where the only assumption on $\{\theta_j\}_j$ is that they are square summable.

Let $M = n^{1/(2r+1)}$ and construct $\theta(\iota) = (\iota_j/\sqrt{M})$ where $\iota_j \in \{\pm 1\}$ for $M \leq j \leq 2M$ and $\iota_j = 0$ otherwise. Easy to check that all $\theta(\iota)$ are square summable there exists a hypothesis

class $\mathcal{I}$ with $\log|\mathcal{I}| \gtrsim M$, such that any pairs of $\iota, \iota' \in \mathcal{I}$ have at least $\Omega(M)$ different signs. Let $f = \sum_{j=1}^{\infty} \alpha_j \psi_j$ such that $\alpha_j = \pm j^{-r}$ with half probabilities, and they are independent. Clearly $\mathbb{E}[\alpha_j^2] = s_j \lesssim j^{-2r}$ and $\mathbb{E}[\alpha_j \alpha_k] = 0$ for $j \neq k$. Then

$$n \times \mathbb{E}[\langle f, \theta(\iota) - \theta(\iota')\rangle^2] \approx \mathrm{KL}(P_\iota^n \| P_{\iota'}^n) \approx \sum_{j=M}^{2M} n s_j |\theta(\iota)_j - \theta(\iota')_j|^2 \asymp \sum_{j=M}^{2M} \frac{n s_j}{M}$$
$$\approx n M^{-2r} = n^{\frac{1}{2r+1}} \asymp \log|\mathcal{I}|.$$

Tsybakov's Master theorem (or Fano's inequality) then shows that minimax rate on MSE error is $n^{-2r/(2r+1)}$.

3.2 Minimax lower bound on honest CIs

Remark 3.1. If we define the ℓ_∞ error between θ, θ' as

$$d(\theta, \theta') := \sup_f \big|\langle f, \theta - \theta'\rangle|\big|$$

where supreme is taken over the support of the distribution on f, the it is easy to see that

$$d(\theta, \theta') = \sum_{j=M}^{2M} j^{-r} |\theta_j - \theta_j'| \approx M \cdot M^{-r}/\sqrt{M} = M^{1/2-r} = \left(n^{\frac{1}{2r+1}}\right)^{\frac{1}{2}-r} = n^{-\frac{2r-1}{2(2r+1)}}$$

whose square is $n^{-(2r-1)/(2r+1)}$. This should hint on a lower bound of honest and adaptive CIs.

4 Extensions

In addition to potential extensions to infinite action space and/or margin conditions, consider also the following model selection problem. We have L possible RKHS with decay rates $r_1, r_2, r_3, \cdots$. The actual β_0 belongs to at least one particular RKHS, but not all of them. Can we adapt to the best rate, without knowing which RKHS?

Remark 4.1. If we know the RKHS, adapting to r isn't a particularly difficult question because we observe T approximately. [2] already did this.

Remark 4.2. This problem is impossible in contextual nonparametric bandit or X-armed bandit settings without additional conditions: see [4, 5]. Self-similarity type conditions must be imposed.

4.1 The idea

Let $\mathcal{H}_1 \supseteq \cdots \supseteq \mathcal{H}_L$ and suppose $r_1 < r_2 < \cdots < r_L$. Suppose we have a sequence of estimates $\widehat{\beta}_\ell$ and CIs $\Delta_\ell(\cdot)$, obtained on the same set of data. We know the following: if $\beta_0 \in \mathcal{H}_{\ell^*}$, then with high probability

$$\big|\langle v, \widehat{\beta}_\ell - \beta_0\rangle\big| \leq \Delta_\ell(v), \;\; \mathbb{E}[\Delta_\ell(v)^2] \lesssim n^{-2r_\ell/(2r_\ell+1)}, \qquad \forall \ell \leq \ell^*.$$

Remark 4.3. If the same training data is used, and $r_1 < r_2 < \cdots < r_L$, it also holds that $\Delta_\ell(v) \leq \Delta_{\ell'}(v)$ for any $\ell \leq \ell'$ and $v \in L^2([0,1])$. This is true for the fix-λ model since $\lambda_\ell > \lambda_{\ell'}$, even if variable CI is used on the variance part. But maybe less so for the anisotropic KRR approach.

Method. Suppose we only have two arms. Each model ℓ is telling us one of the three things: arm 0 is optimal, arm 1 is optimal, or indifference. Go from the $\ell = 1, 2, \cdots$ and stop if a decision is made. Use that decision.

Analysis. If we stop at $\widehat{\ell} \leq \ell^*$, then the decision is correct and the analysis proceeds as usual. If we stop at $\widehat{\ell} > \ell^*$, it means the decision of ℓ^* is "indifferent" and therefore regret on x is upper bounded by $\Delta_{\ell^*}(x)$. We only need to make sure that each action gets enough data. This is done by introducing some auxiliary data (e.g. pure exploration) to make sure $\Delta_\ell(x) = o(1)$ for all ℓ and x, and strengthening the dominance condition in the the above section to apply on all data with a positive margin.

5 High-order terms

Remark 5.1. If we do assume the population covariance T is known, it is probably a better idea to use

$$(T + \lambda I)^{-1} \frac{1}{n} \sum_{i=1}^{n} y_i L_{K^{1/2}} f_i$$

as the estimator, which might be easier to analyze.

Remark 5.2. In the anisotropic KRR method, if we insist on not knowing T, we have to adjust the estimator to

$$(T_n + \Lambda_n)^{-1} \frac{1}{n} \sum_{i=1}^{n} y_i L_{K^{1/2}} f_i$$

where $\Lambda_n = \sum_{j=1}^{\infty} \lambda_j \widehat{\psi}_j \otimes \widehat{\psi}_j$, where $\{\widehat{\psi}_j\}$ are eigenfunctions of T_n. We may then impose all assumptions on T_n and $\{\widehat{\psi}_j\}$ instead. We may still need to analyze $T - T_n$ because $\mathbb{E}[\Delta(\widetilde{v})^2]$ must be on the population level (new data are different from training data for sure.)

References

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